

EXAM #1

DYUT DAS

1. WHAT ARE THE INDIVIDUAL LAYERS IN A TRISO PARTICLE? WHAT PURPOSE DOES EACH LAYER HAVE?

9/10

Fuel Kernel: FUEL, PROVIDES FISSIONS/~~FUEL~~ SOURCE, UO_2 OR UCO

Buffer Layer: PROVIDES POROUS LAYER FOR FP GASES EXPANSION TO PROTECT/PREVENT SiC LAYER DAMAGE OR FAILURE

IPyC Layer: PROVIDES PROTECTIVE LAYER TO SiC LAYER FROM FP GASES EXPANSION + IRRADIATION SWELLING/SHRINKAGE

SiC Layer: PROVIDES PRIMARY PRESSURE BOUNDARY FOR FUEL KERNEL TO PREVENT RELEASE OF FPs.

OPyC Layer: PROTECTS SiC LAYER AND PROVIDES SUPPORT FOR IRRADIATION EFFECTS
- IPyC/OPyC, SiC into compression

2. WHAT ARE THE TRADE-OFFS BETWEEN CARBIDE AND OXIDE FUEL KERNELS?

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MAIN TRADE OFF IS THE PRODUCTION OF CO AND CO_2 IS MINIMIZED W/ CARBIDE KERNELS. THIS PUTS LESS STRESS ON OTHER LAYERS OF FUEL KERNEL. ADDITIONALLY, OXIDE KERNELS MAY PRODUCE MORE INSOLUBLE/SOLUBLE OXIDE FPs WHICH MAY LEAD TO MORE FP BUILD UP INSIDE FUEL KERNEL. - kind of... oxides retain FPs better

3. DISCUSS FISSION PRODUCT CHEMISTRY IN THE FUEL KERNEL

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FISSION PRODUCTS CAN BE SOLUBLE OXIDES, INSOLUBLE OXIDES, METALS, NERVE GASES, PRECIPITATES. THE RATE OF PRODUCTION IS DRIVEN BY THE FISSION OF ^{235}U NUCLEI, DECAY OF UNSTABLE NUCLEIDES, AND LOSS THROUGH BURNUP. FISSION PRODUCTS WILL REACT WITH OXIDES TO CREATE SOLUBLE OXIDES, INSOLUBLE OXIDES, METAL OXIDES WHICH CONTRIBUTE TO FP BUILD UP INSIDE THE FUEL KERNEL.

- needed more here

4. HOW DOES THE NATURE OF IRRADIATION DAMAGE ACCUMULATION CHANGE W/ IRRADIATION TEMPERATURE IN SiC? HOW DOES IT IMPACT SWELLING?

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At $< 423\text{K}$, SiC REACHES AMORPHIZATION REGIME.

At $\sim 423\text{K} \rightarrow 1273\text{K}$, SiC IS IN POINT DEFECT SWELLING REGIME. AS TEMPERATURE $\uparrow$,

DOSE SATURATION $\downarrow$ AS MORE POINT DEFECTS FORM. - Swelling saturates not dose

At $\sim 1273\text{K}$, TRANSITIONS TO VOID SWELLING REGIME $\rightarrow$ THERE IS NO SATURATION POINT

At HIGHER TEMPERATURES, SWELLING MAY CAUSE SiC FAILURE

THE IMPACT OF IRRADIATION SWELLING IS REDUCED BY IRRADIATION CRACKS.

- I think you implied saturation, rather than stating it...

5. HOW DOES THERMAL CONDUCTIVITY CHANGE IN SiC W/ IRRADIATION? WHAT ARE THE PRIMARY PHONON SCATTERERS?

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THERMAL CONDUCTIVITY IS DIRECTLY PROPORTIONAL TO IRRADIATION IN THE POINT DEFECT SWELLING REGIME BUT ONCE IN THE VOID SWELLING REGIME, THERE IS NO IMPACT TO THERMAL CONDUCTIVITY. PRIMARY PHONON SCATTERERS ARE INTERSTITIAL CRACKS, POINT DEFECTS, AND SMALLER VOIDS.

- how is it proportional? be explicit

- voids still impact K_{th}

6. HOW DOES GRAPHITE DIMENSIONALLY CHANGE UNDER IRRADIATION? WHY DOES THIS BEHAVIOR HAPPEN?

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GRAPHITE WILL 'SHRINK' UNTIL IT'S 'TURNAROUND' POINT AT WHICH POINT SWELLING WILL COMMENCE. THIS OCCURS BECAUSE IT BEHAVES ANISOTROPICALLY AT LOWER TEMPERATURES. - anisotropic at all T

THE 'C' PLANE WILL EXPAND WHILE THE 'A' PLANE WILL COMPRESS. THIS WILL CONTINUE UNTIL NO MORE 'GAPS' CAN COMPRESS/BE FILLED. THIS IS THE TURNAROUND POINT AND AT THIS POINT GRAPHITE WILL START TO SWELL. THIS OCCURS AT APPROXIMATELY 2000K. - why?

- interstitials + vacancies? CTE + Mrozowski cracks?

7. LIST AND EXPLAIN THREE TYPES OF FAILURE MECHANISMS FOR TRISO PARTICLES.

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1. ^{over pressure} PRESSURE BOUNDARY FAILURE - FP, CO, AND CO₂ GAS BUILD UP CAUSES PRESSURE BOUNDARY OR SiC LAYER FAILURE AND RELEASE OF FP

2. ^{- more often fabrication issues} IPYC AND SiC DEBONDING - STRESSES + STRAINS ON IPYC LAYER + SiC LAYER CAUSE THE TWO LAYERS TO COME APART AND INTRODUCE FRAGILES IN THE LAYERS.

3. SiC NON-RETENTION - SiC LAYER IRRADIATION RESULTS IN SEVERE DEGRADATION OF THE LAYER AND IT FAILS TO PERFORM ITS FUNCTION AS A PRESSURE BOUNDARY FOR FUEL KERNEL.

8. PROVIDE ONE EXAMPLE OF AN ADVANCED TRISO CONCEPT AND EXPLAIN WHY IT IS OF INTEREST

5/5

ZrC INSTEAD SiC LAYER → PROVIDES THE ABILITY TO OPERATE AT HIGHER TEMPERATURES AND IS ABLE TO PREVENT Pd MIGRATION. IT WILL AFFORD THE OPPORTUNITY TO GATHER MORE DATA FOR OTHER ASPECTS OF ITER + TRISO FUELS AT HIGHER TEMPERATURES.

9. WHAT ARE THE CONCERNS FOR INTERMEDIATE HX MATERIALS IN A He ENVIRONMENT? DISCUSS CORROSION OF THESE MATERIALS.

6/10

He COOLANT HAS NATURAL OCCURRENCE OF VARIOUS IMPURITIES WHICH REACT WITH VARIOUS REACTOR COMPONENTS WHICH CAUSE OXIDATION + CORROSION. FOR INTERMEDIATELY, IT IS THE MAIN COMPONENT THAT HAS THE MOST IMPORTANT INTERACTION W/ He WITH COOLANT RUNNING THROUGH THE HX @ ~100 m/s. INCREASED CORROSION WILL ↓ HEAT XFER CAPABILITIES AND ↑ INTERNAL FUEL TEMPERATURES POTENTIALLY RESULTING IN MORE FUEL KERNEL FAILURES. THE USE OF NICKEL-BASIS ALLOYS REDUCES THE

AMOUNT OF CORROSION $\rightarrow$ SPECIFICALLY ALLOY-617. AT HIGH TEMPERATURES, THE IMPACT OF CORROSION IS LESS AND INCREASED THERMAL STRESSES MAY CAUSE FAILURE OF REACTOR MATERIALS. ALLOY 617 IS TESTED UP TO 750 K AND MAY BE SUCCESSFUL UP TO 950 K.

- needed a lot more here, Carburization + decarburization.

Oxygen potential? Carbide + oxide layer ...